File Edit View Navigate Code Refactor Run Tools VCS Window Help 04-SparkSchemaDemo - SparkSchemaDemo.py - PyCharm

04-SparkSchemaDemo > SparkSchemaDemo.py

Project

- 04-SparkSchemaDemo C:\Users\prashant\Documents\GitHub\S
  - app-logs
  - data
  - lib
  - log4j.properties
  - SparkSchemaDemo.py
- External Libraries
- Scratches and Consoles

SparkSchemaDemo.py

```
        .option("header", "true") \
        .option("inferSchema", "true") \
        .load("data/flight*.csv")

    flightTimeCsvDF.show(5)
    logger.info("CSV Schema:" + flightTimeCsvDF.schema.simpleString())

if __name__ == "__main__"
```

Run: SparkSchemaDemo

```
C:\Users\prashant\anaconda3\envs\pyspark\python.exe C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/04-SparkSchemaDemo/SparkSchemaDemo.py
20/06/14 08:42:37 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
+---------+----------+-----------------+------+---------------+----+-------------+-----------+--------+---------+-------+-----------+--------+---------+--------+
| FL_DATE|OP_CARRIER|OP_CARRIER_FL_NUM|ORIGIN|ORIGIN_CITY_NAME|DEST|DEST_CITY_NAME|CRS_DEP_TIME|DEP_TIME|WHEELS_ON|TAXI_IN|CRS_ARR_TIME|ARR_TIME|CANCELLED|DISTANCE|
+---------+----------+-----------------+------+---------------+----+-------------+-----------+--------+---------+-------+-----------+--------+---------+--------+
|1/1/2000|        DL|             1451|   BOS|     Boston, MA| ATL|   Atlanta, GA|       1115|    1113|     1343|      5|       1400|    1348|        0|     946|
|1/1/2000|        DL|             1479|   BOS|     Boston, MA| ATL|   Atlanta, GA|       1315|    1311|     1536|      7|       1559|    1543|        0|     946|
|1/1/2000|        DL|             1857|   BOS|     Boston, MA| ATL|   Atlanta, GA|       1415|    1414|     1642|      9|       1721|    1651|        0|     946|
|1/1/2000|        DL|             1997|   BOS|     Boston, MA| ATL|   Atlanta, GA|       1715|    1720|     1955|     10|       2013|    2005|        0|     946|
|1/1/2000|        DL|             2065|   BOS|     Boston, MA| ATL|   Atlanta, GA|       2015|    2010|     2230|     10|       2300|    2240|        0|     946|
+---------+----------+-----------------+------+---------------+----+-------------+-----------+--------+---------+-------+-----------+--------+---------+--------+
only showing top 5 rows

20/06/14 08:42:49 INFO SparkSchemaDemo: CSV Schema:struct<FL_DATE:string,OP_CARRIER:string,OP_CARRIER_FL_NUM:int,ORIGIN:string,ORIGIN_CITY_NAME:string,DEST:string,DEST_CITY_NAME:st

Process finished with exit code 0
```

9: Git | 6: TODO | 4: Run | Terminal | Python Console | Event Log

Spark DataFrameReader

Schema for DataFrame

1. Explicit
2. Implicit

kSchemaDemo 〉 SparkSchemaDemo.py

Project

04-SparkSchemaDemo C:\Users\prashant\Documents\GitHub\S

- app-logs
- data
- lib
- log4j.properties
- SparkSchemaDemo.py

External Libraries

Scratches and Consoles

SparkSchemaDemo.py

```
flightTimeJsonDF.show(5)
logger.info("JSON Schema:" + flightTimeJsonDF.schema.simpleString())
```

SparkSchemaDemo

```
GA|        1715|    1720|     1955|    10|        2013|    2005|        0|     946|
GA|        2015|    2010|     2230|    10|        2300|    2240|        0|     946|
---+------------+--------+---------+------+------------+--------+---------+--------+


RIER:string,OP_CARRIER_FL_NUM:int,ORIGIN:string,ORIGIN_CITY_NAME:string,DEST:string,DEST_CITY_NAME:string,CRS_DEP_TIME:int,DEP_TIME:int,WHEELS_ON:int,TAXI_IN:int,CRS_ARR_TIME:int,A
----+--------+----------+-----------------+------+---------------+-------+---------+
ANCE| FL_DATE|OP_CARRIER|OP_CARRIER_FL_NUM|ORIGIN|ORIGIN_CITY_NAME|TAXI_IN|WHEELS_ON|
----+--------+----------+-----------------+------+---------------+-------+---------+
 946|1/1/2000|        DL|             1451|   BOS|      Boston, MA|      5|     1343|
 946|1/1/2000|        DL|             1479|   BOS|      Boston, MA|      7|     1536|
 946|1/1/2000|        DL|             1857|   BOS|      Boston, MA|      9|     1642|
 946|1/1/2000|        DL|             1997|   BOS|      Boston, MA|     10|     1955|
 946|1/1/2000|        DL|             2065|   BOS|      Boston, MA|     10|     2230|
----+--------+----------+-----------------+------+---------------+-------+---------+


ELLED:bigint,CRS_ARR_TIME:bigint,CRS_DEP_TIME:bigint,DEP_TIME:bigint,DEST:string,DEST_CITY_NAME:string,DISTANCE:bigint,FL_DATE:string,OP_CARRIER:string,OP_CARRIER_FL_NUM:bigint,ORI
```

Git | TODO | Run | Terminal | Python Console

Event Log

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SparkSchemaDemo > SparkSchemaDemo.py

Project

04-SparkSchemaDemo C:\Users\prashant\Documents\GitHub\S
- app-logs
- data
  - flight-time.csv
  - flight-time.json
  - flight-time.parquet
- lib
- log4j.properties
- SparkSchemaDemo.py
- External Libraries
- Scratches and Consoles

SparkSchemaDemo.py

```python
from pyspark.sql import SparkSession

from lib.logger import Log4j

if __name__ == "__main__":
    spark = SparkSession \
        .builder \
        .master("local[3]") \
        .appName("SparkSchemaDemo") \
```

Run: SparkSchemaDemo

```
|   2005|      0|      2013|      1715|   1720| ATL|   Atlanta, GA|    946|1/1/2000|      DL|      1997|  BOS|    Boston, MA|    10|   1955|
|   2240|      0|      2300|      2015|   2010| ATL|   Atlanta, GA|    946|1/1/2000|      DL|      2065|  BOS|    Boston, MA|    10|   2230|
+--------+---------+----------+----------+-------+----+--------------+-------+--------+--------+----------+-----+--------------+------+-------+
only showing top 5 rows

20/06/14 09:45:17 INFO SparkSchemaDemo: JSON Schema:struct<ARR_TIME:bigint,CANCELLED:bigint,CRS_ARR_TIME:bigint,CRS_DEP_TIME:bigint,DEP_TIME:bigint,DEST:string,DEST_CITY_NAME:strin
+----------+----------+----------------+------+----------------+----+--------------+------------+--------+---------+-------+------------+--------+---------+--------+
|  FL_DATE|OP_CARRIER|OP_CARRIER_FL_NUM|ORIGIN|ORIGIN_CITY_NAME|DEST|DEST_CITY_NAME|CRS_DEP_TIME|DEP_TIME|WHEELS_ON|TAXI_IN|CRS_ARR_TIME|ARR_TIME|CANCELLED|DISTANCE|
+----------+----------+----------------+------+----------------+----+--------------+------------+--------+---------+-------+------------+--------+---------+--------+
|2000-01-01|        DL|            1451|   BOS|      Boston, MA| ATL|   Atlanta, GA|        1115|    1113|     1343|      5|        1400|    1348|        0|     946|
|2000-01-01|        DL|            1479|   BOS|      Boston, MA| ATL|   Atlanta, GA|        1315|    1311|     1536|      7|        1559|    1543|        0|     946|
|2000-01-01|        DL|            1857|   BOS|      Boston, MA| ATL|   Atlanta, GA|        1415|    1414|     1642|      9|        1721|    1651|        0|     946|
|2000-01-01|        DL|            1997|   BOS|      Boston, MA| ATL|   Atlanta, GA|        1715|    1720|     1955|     10|        2013|    2005|        0|     946|
|2000-01-01|        DL|            2065|   BOS|      Boston, MA| ATL|   Atlanta, GA|        2015|    2010|     2230|     10|        2300|    2240|        0|     946|
+----------+----------+----------------+------+----------------+----+--------------+------------+--------+---------+-------+------------+--------+---------+--------+
only showing top 5 rows

20/06/14 09:45:19 INFO SparkSchemaDemo: Parquet Schema:struct<FL_DATE:date,OP_CARRIER:string,OP_CARRIER_FL_NUM:int,ORIGIN:string,ORIGIN_CITY_NAME:string,DEST:string,DEST_CITY_NAME:

Process finished with exit code 0
```

9: Git | 6: TODO | 4: Run | Terminal | Python Console | Event Log

PEP 8: W391 blank line at end of file

36:5 CRLF UTF-8 4 spaces Python 3.7 (pyspark) master

Apache Spark

Data Frame Schema

udemy

Spark DataFrame Schema

Spark Data Types : https://spark.apache.org/docs/latest/api/python/pyspark.sql.html#module-pyspark.sql.types

S.No.	Spark Types	Scala Types	Python Types
1.	IntegerType	Int	Int
2.	LongType	Long	long
3.	FloatType	Float	Float
4.	DoubleType	Double	Float
5.	StringType	String	String
6.	DateType	java.sql.Date	datetime.date
8.	TimestampType	java.sql.Timestamp	datetime.datetime
9.	ArrayType	scala.collection.Seq	List, tuple, or array
10.	MapType	scala.collection.Map	dict

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How to define Schema?

Udemy

Spark DataFrame Schema

Spark Schema

1. Programmatically
2. Using DDL String

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www.learningjournal.guru

```
flightSchemaStruct = StructType([
    StructField("FL_DATE", DateType()),
    StructField("OP_CARRIER", StringType()),
    StructField("OP_CARRIER_FL_NUM", IntegerType()),
    StructField("ORIGIN", StringType()),
    StructField("ORIGIN_CITY_NAME", StringType()),
    StructField("DEST", StringType()),
    StructField("DEST_CITY_NAME", StringType()),
    StructField("CRS_DEP_TIME", IntegerType()),
    StructField("DEP_TIME", IntegerType()),
    StructField("WHEELS_ON", IntegerType()),
    StructField("TAXI_IN", IntegerType()),
    StructField("CRS_ARR_TIME", IntegerType()),
    StructField("ARR_TIME", IntegerType()),
    StructField("CANCELLED", IntegerType()),
    StructField("DISTANCE", IntegerType())
])

flightTimeCsvDF = spark.read \
    .format("csv") \
    .option("header", "true") \
    .schema(flightSchemaStruct) \
    .option("mode", "FAILFAST") \
    .load("data/flight*.csv")

flightTimeCsvDF.show(5)
```

```
flightTimeCsvDF = spark.read \
    .format("csv") \
    .option("header", "true") \
    .schema(flightSchemaStruct) \
    .option("mode", "FAILFAST") \
    .option("dateFormat", "M/d/y") \
    .load("data/flight*.csv")

flightTimeCsvDF.show(5)
logger.info("CSV Schema:" + flightTimeCsvDF.schema.simpleString())

flightTimeJsonDF = spark.read \
    .format("json") \
    .load("data/flight*.json")

flightTimeJsonDF.show(5)
logger.info("JSON Schema:" + flightTimeJsonDF.schema.simpleString())
```

```
25          StructField("WHEELS_ON", IntegerType()),
26          StructField("TAXI_IN", IntegerType()),
27          StructField("CRS_ARR_TIME", IntegerType()),
28          StructField("ARR_TIME", IntegerType()),
29          StructField("CANCELLED", IntegerType()),
30          StructField("DISTANCE", IntegerType())
31      ])
32
33      flightSchemaDDL = """FL_DATE DATE, OP_CARRIER STRING, OP_CARRIER_FL_NUM INT, ORIGIN STRING,
34            ORIGIN_CITY_NAME STRING, DEST STRING, DEST_CITY_NAME STRING, CRS_DEP_TIME INT, DEP_TIME INT,
35            WHEELS_ON INT, TAXI_IN INT, CRS_ARR_TIME INT, ARR_TIME INT, CANCELLED INT, DISTANCE INT"""
36
37      flightTimeCsvDF = spark.read \
38          .format("csv") \
39          .option("header", "true") \
40          .schema(flightSchemaStruct) \
41          .option("mode", "FAILFAST") \
42          .option("dateFormat", "M/d/y") \
43          .load("data/flight*.csv")
44
45      flightTimeCsvDF.show(5)
46      logger.info("CSV Schema:" + flightTimeCsvDF.schema.simpleString())
47
```